

Theorem 4: Let $X_1, X_2, \dots, X_n$ be independent random variables such that $E|X_i| < \infty$, $i = 1, 2, \dots, n$. Then $E\left(\prod_{i=1}^n X_i\right)$ exists and

$$E\left(\prod_{i=1}^n X_i\right) = \prod_{i=1}^n E(X_i). \quad (2)$$

Remark 3: Let X and Y be independent and $g_1(\cdot)$ and $g_2(\cdot)$ be Borel-measurable functions. Then $g_1(X)$ and $g_2(Y)$ are independent. If $E[g_1(X)]$, $E[g_2(Y)]$, and $E[g_1(X)g_2(Y)]$ exist, then

$$E[g_1(X), g_2(Y)] = E[g_1(X)]E[g_2(Y)]. \quad (3)$$

Theorem 5: Let $X_1, X_2, \dots, X_n$ be random variables with $E|X_i|^2 < \infty$, $i = 1, 2, \dots, n$. Let $a_1, a_2, \dots, a_n$ be real numbers and write $S = \sum_{i=1}^n a_i X_i$. Then the variance of S exists and is given by

$$Var(S) = \sum_{i=1}^n a_i^2 Var(X_i) + \sum_{i \neq j=1}^n a_i a_j Cov(X_i, X_j). \quad (4)$$

If in particular $X_1, X_2, \dots, X_n$ are such that $Cov(X_i, X_j) = 0$ for $i, j = 1, 2, \dots, n$, $i \neq j$, then

$$Var(S) = \sum_{i=1}^n a_i^2 Var(X_i) \quad (5)$$

Proof: We have

$$\begin{aligned} Var(S) &= E\left(\sum_{i=1}^n a_i X_i - \sum_{i=1}^n a_i E(X_i)\right)^2 \\ &= E\left[\sum_{i=1}^n a_i^2 (X_i - E(X_i))^2 + \sum_{i \neq j=1}^n a_i a_j (X_i - E(X_i))(X_j - E(X_j))\right] \\ &= \sum_{i=1}^n a_i^2 (X_i - E(X_i))^2 + \sum_{i \neq j=1}^n a_i a_j E[(X_i - E(X_i))(X_j - E(X_j))]. \end{aligned}$$

If $Cov(X_i, X_j) = 0$ for $i, j = 1, 2, \dots, n$, $i \neq j$, the second term on the right side of (4) vanishes, and we have (5).

Corollary: Let $Var(X_i) = \sigma^2$, $i = 1, 2, \dots, n$. Then

$$Var\left(\sum_{i=1}^n a_i X_i\right) = \sigma^2 \sum_{i=1}^n a_i^2 + \rho \sigma^2 \sum_{i \neq j=1}^n a_i a_j,$$

where ρ is the correlation coefficient between X_i and X_j , $i \neq j$. In particular,

$$Var\left(\sum_{i=1}^n \frac{X_i}{n}\right) = \frac{\sigma^2}{n} + \frac{n-1}{n} \rho \sigma^2.$$

Corollary: If $X_1, X_2, \dots, X_n$ are uncorrelated, then

$$Var\left(\sum_{i=1}^n a_i X_i\right) = \sigma^2 \sum_{i=1}^n a_i^2,$$

and

$$Var\left(\sum_{i=1}^n \frac{X_i}{n}\right) = \frac{\sigma^2}{n}.$$

Theorem 6: Let $X_1, X_2, \dots, X_n$ be iid random variables with common variance σ^2 . Also let $a_1, a_2, \dots, a_n$ be real numbers such that $\sum_{i=1}^n a_i = 1$, and let $S = \sum_{i=1}^n a_i X_i$. Then the variance of S is least if we choose $a_i = \frac{1}{n}$, $i = 1, 2, \dots, n$.

Proof: We have

$$Var(S) = \sigma^2 \sum_{i=1}^n a_i^2,$$

which is least if and only if we choose the a_i 's so that $\sum_{i=1}^n a_i^2$ is smallest, subject to the condition

$\sum_{i=1}^n a_i = 1$. We have

$$\sum_{i=1}^n a_i^2 = \sum_{i=1}^n \left(a_i - \frac{1}{n} + \frac{1}{n}\right)^2 = \sum_{i=1}^n \left(a_i - \frac{1}{n}\right)^2 + \frac{2}{n} \sum_{i=1}^n \left(a_i - \frac{1}{n}\right) + \frac{1}{n} = \sum_{i=1}^n \left(a_i - \frac{1}{n}\right)^2 + \frac{1}{n},$$

which is minimized for the choice $a_i = 1/n$, $i = 1, 2, \dots, n$.

Conditional Expectation: Let X and Y be random variables defined on a probability space $(\Omega, \mathcal{F}, P)$, and let h be a Borel-measurable function. Then the conditional expectation of $h(X)$, given Y , written as $E\{h(X)|Y\}$, is a random variable that takes the value $E\{h(X)|y\}$, defined by

$$E\{h(X)|y\} = \begin{cases} \sum_{-\infty}^{\infty} h(x)P\{X = x|Y = y\}, & \text{if } (X, Y) \text{ is discrete and } P\{Y = y\} > 0, \\ \int_{-\infty}^{\infty} h(x)f_{X|Y}(x|y)dx, & \text{if } (X, Y) \text{ is continuous and } f_2(y) > 0, \end{cases} \quad (6)$$

where the random variable Y assumes the value y .

A similar definition can be given for the conditional expectation $E\{h(Y)|X\}$.